When pooling costs a decision: probability calibration with Calibre*

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Abstract

Isotonic calibration fits flat regions: runs of scores that receive one probability. A flat region costs a decision only when the cases it pools fall on both sides of the decision's threshold, and a ranking only when its ties are broken at random. We make that condition precise, separate it from the cost of acting on a misestimated probability, and implement the comparison in Calibre, a Python package of calibration maps, proper-score diagnostics, and decision evaluation. Evidence comes from an exact known-risk simulation and a pre-registered benchmark of 7 synthetic designs and one real dataset over 30 seeds. Flat regions are cheap for cost-threshold decisions in these designs: a rule that knew the true probability would gain a median of 0.47 per 1,000 cases over isotonic regression, and choosing a calibrator by validation payoff instead of validation Brier score did no better (8 of 130 intervals above zero and 11 below). They matter more for allocation: random tie-breaking costs up to 1.7 successes per 1,000 cases, which breaking ties by the original score recovers exactly. A method that builds the decision thresholds into the fit lost in every configuration. The practical rule is to choose a calibrator on a proper score, break its ties by the original score, and check whether its flat regions straddle the thresholds in use.

1 Introduction

A model can rank cases well without telling a user whether acting on any of them is worth the cost. Suppose an inspection costs c, finding a defect is worth $b > 0$, and Y indicates a defect. Inspecting pays $bY - c$, so a user who knows the defect probability p inspects when $p > t$, where $t = c/b$ is the cost ratio. Ranking cases does not locate that boundary. Calibration does: it estimates the probability that goes with each score, so that a reported probability Q can be compared with t.

The standard calibrator, isotonic regression, fits a non-decreasing step function. Its steps are flat regions: runs of adjacent scores that receive the same probability. Pooling scores this way stabilizes the estimate but discards distinctions, and whether the discarded distinctions mattered depends on the decision. A flat region that lies entirely above or entirely below t costs nothing; every case in it gets the right action anyway. One that straddles t assigns a single action to cases that should get different ones. Calibration error cannot see the difference. Reporting the population event rate for every case is perfectly calibrated and useless for targeting.

This paper makes the condition precise, measures how often it binds, and supplies software to check it. Section 2 shows that pooling loses value at a threshold exactly when a flat region holds cases on both sides of it, and separates that loss from the cost of acting on a misestimated probability. For allocation under a fixed capacity, ties are the only harm a monotone map can do, and breaking them by the original score removes it. Section 3 treats the alternative calibrators as different assumptions about flat regions. Section 5 reports the evidence.

The main empirical finding is that flat regions are cheap for cost-threshold decisions in the designs we study and costly for allocation only through tie-breaking. Choosing a calibrator by validation payoff, the natural decision-aware procedure, did no better than choosing it by validation Brier score. A

*Software and replication materials: `https://github.com/finite-sample/calibre`.

calibrator that builds the decision thresholds into the fit lost to isotonic regression in every configuration. These are negative results for decision-aware tuning, and they translate into a simple procedure, set out in Section 6.

The theory specializes established results on the value of information and grouping loss (Blackwell, 1953; Tasche, 2021; Guo et al., 2026), and original-score tie-breaking is known (Menon et al., 2012). Appendix C records which claims are established and which are specializations. The contribution is the assembled test, an implementation of it with explicit data roles, and evidence on the size of the costs.

2 When a flat region costs a decision

Fix a fitted calibration map g and write $Q = g(S)$ for the probability it reports for a case with score S. The true risk is $p = P(Y = 1 \mid S)$, and the risk recoverable from the report is $r = E[p \mid Q]$. Within a flat region, Q is constant while p may vary; r is the average risk of the cases the region pools. Actions do not affect Y.

An oracle who knows the population law and sees S earns $V_t(S) = E[(p - t)_+]$ per case, in benefit-normalized units. Seeing only Q, the oracle earns $V_t(Q) = E[(r - t)_+]$. A user who acts when $Q > t$ earns $U_t(Q) = E[(p - t)\mathbf{1}\{Q > t\}]$. The shortfall from the best possible payoff splits in two:

$$V_t(S) - U_t(Q) = \underbrace{V_t(S) - V_t(Q)}_{\text{information lost through pooling}} + \underbrace{V_t(Q) - U_t(Q)}_{\text{error from the reported level}} .$$

Both terms are nonnegative. The first compares information sets; the second compares uses of the same information.

The first term is zero exactly when no report group holds cases strictly on both sides of t (Proposition 1, Appendix A). Variation entirely on one side of the threshold does not change the optimal action, so it costs nothing to pool. Figure 1 shows the condition with two equally common groups whose risks are $0.3 \pm d$: pooling them costs value only for thresholds between the two risks. Across all thresholds weighted uniformly, the first term integrates to half the squared grouping loss $E[(p - r)^2]$, and the second to half the miscalibration $E[(r - Q)^2]$; together they are half the Brier score's excess over the true probabilities (Ehm et al., 2016). A calibrated report ($Q = r$) is therefore usable at face value relative to its own information. Replacing the score without loss at every threshold also requires $p = r$.

Allocation under a fixed capacity is a different decision. If exactly k equal-sized slots are available and each success is worth the same, the user selects the k highest risks, and a correctly ordered score suffices even when its levels are wrong. A non-decreasing map cannot improve that order; it can only tie cases the score distinguished. Ranking by Q and breaking its ties by S reproduces the original-score ranking exactly, at every capacity (Proposition 4, Appendix B). When the original order is itself wrong, as with nonmonotone risk, restoring it does not help.

3 Calibration maps as assumptions about flat regions

Every calibrator in Calibre answers the same question differently: where should the fitted map be flat? Table 1 lists them. For scores s_i with binary outcomes y_i and weights w_i, tied scores are pooled first: for each unique score x_j the target $\bar{y}_j$ is the weighted outcome mean and W_j the total weight. A deterministic map cannot separate observations with exactly the same score.

Isotonic regression minimizes $\sum_j W_j(\bar{y}_j - z_j)^2$ subject to $z_1 \le \cdots \le z_m$, solved in linear time by the pool-adjacent-violators algorithm (PAVA) (Ayer et al., 1955). Its flat regions are features of the constrained fit; they do not establish that the pooled cases share a risk. For binary labels, the PAVA solution optimizes every regular binary proper scoring rule under the ordering constraint on the fitting

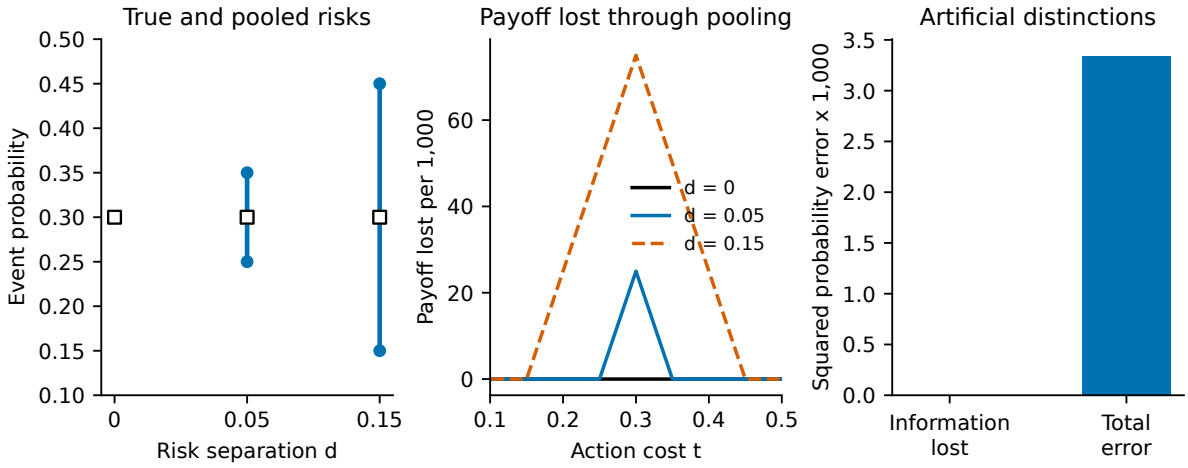


Figure 1: Distinctness is valuable only through decisions. Left: two equally common groups have risks $0.3 \pm d$; squares mark their pooled forecast. Middle: exact optimal payoff lost through pooling, per 1,000 cases, as the cost ratio varies. Right: a constant-risk population has 1,000 forecasts evenly spaced from 0.2 to 0.4; the artificial distinctions add probability error without risk information. These constructed examples have no sampling uncertainty.

Table 1: Calibration maps in Calibre. "Linear" means interpolation between fitted anchors, with constant extrapolation beyond them. Non-decreasing maps can contain flat regions. Neither nearly isotonic regression nor a negative increment bound guarantees monotonicity.

Method	Fit	Shape and prediction	Tuning
Isotonic	Weighted least squares under ordering	Non-decreasing; linear	None
Centered isotonic	Pooled rates at block anchors	Non-decreasing; linear	None
Nearly isotonic	Squared loss plus penalty on decreases	Decreases permitted; linear	Penalty by cross-validation
I-spline	Penalized Bernoulli or squared loss	Non-decreasing spline	Knots and penalty by cross-validation
Relaxed PAVA	Squared loss with adjacent increment bound	Depends on bound; linear	User-specified bound
CDI-ISO	Squared loss with local increment bounds	Decreases possible; step	User supplies thresholds

sample (Brümmer and du Preez, 2013). That is a statement about fitted values, not about out-of-sample decisions. Calibre's isotonic estimator wraps scikit-learn's.

Centered isotonic regression collapses each pooled block to an anchor at its weighted score centroid and interpolates between anchors (Oron and Flournoy, 2017). It removes interior flat regions by assuming risk varies linearly between anchors. It does not minimize the isotonic objective, and the assumption can be wrong.

The I-spline calibrator fits $\ell\{g(s)\} = \theta + \sum_k \delta_k I_k(s)$ with $\delta_k \geq 0$, where the I_k are non-decreasing I-spline basis functions and ℓ is the logit or identity link (Ramsay, 1988). A penalty on successive coefficient differences controls roughness (Eilers and Marx, 1996), and cross-validation selects the knot count and penalty. Flat regions appear only where the fitted increments vanish. When the scores nearly separate the classes, the logistic fit has no finite optimum and approaches a step; Calibre then warns rather than failing.

Nearly isotonic regression replaces the ordering constraint with a penalty $\lambda \sum_j (z_j - z_{j+1})_+$ on decreases, solved along its exact path (Tibshirani et al., 2011). A finite penalty can reverse ranks. Relaxed PAVA imposes $z_{j+1} - z_j \geq L$ with a user-chosen bound: zero recovers isotonic regression, a negative value permits bounded decreases, and a positive value forces separation. A cumulative shift reduces it to one isotonic projection (Appendix D).

CDI-ISO, Calibre's cost- and data-informed variant, applies the decision logic of Section 2 inside the fit. Near user-supplied thresholds it requires an increase between adjacent scores whose event rates differ by more than their sampling uncertainty; elsewhere it permits bounded decreases (Appendix E). Logistic recalibration of the log-odds and temperature scaling never pool; they impose a parametric shape instead (Guo et al., 2017).

Figure 2 shows what these assumptions do to one calibration sample. Isotonic regression and its nearly isotonic relative produce long flat regions; centered isotonic and the spline pass through them.

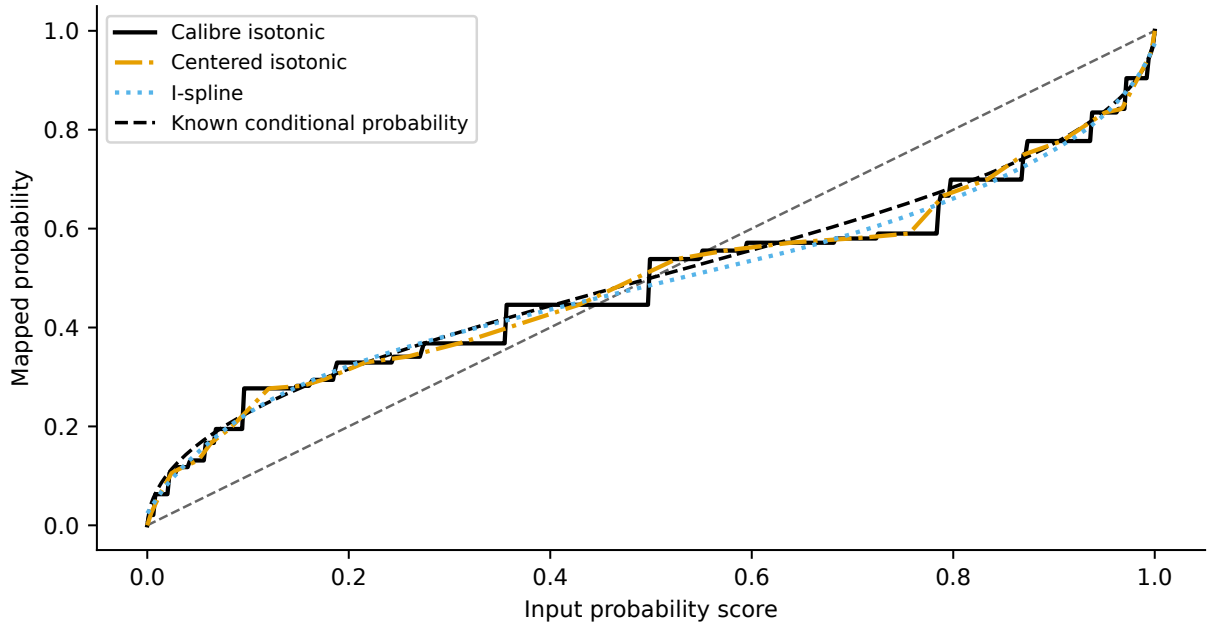


Figure 2: Calibration maps fitted to one calibration sample of the overconfidence design (seed zero). The dashed curve is the true conditional event probability; the diagonal is no recalibration. Curves show pooling and interpolation behavior; they carry no uncertainty bands.

4 Evaluating a calibration procedure for a decision

The object to evaluate is a procedure: a fitted map, the information reported, and the rule for acting on it. Calibre separates three data roles. Calibrators are fitted on scores the base model did not train on. A procedure is selected on a separate validation sample, against a declared objective. The selected procedure is frozen and evaluated once on untouched test data.

The decision module takes a cost ratio or a capacity and returns the mean realized payoff of each candidate procedure on labeled cases, with act-all and act-none baselines for cost decisions and the original-score ranking for capacity. On independent test cases, mean realized payoff estimates a fixed rule's population payoff without knowing any case's true risk. Paired differences compare procedures on the same cases. Capacity ties are integrated out exactly rather than drawn at random, and a tie-breaking score can be supplied. Selection keeps the first highest-payoff candidate; evaluation refuses test identifiers that overlap the validation set. Paired bootstrap intervals resample independent cases and condition on the fitted and selected procedure.

The split between information lost and level error in Section 2 needs the true risk, so it is an oracle quantity. It is available in simulation and not from labels: treating each unique forecast's observed outcome as its risk would give a misleading estimate. Calibre therefore reports it only for known-risk experiments.

The workflow in code, fitting on one independent subset and evaluating on another:

```
import numpy as np

from calibre import CenteredIsotonicCalibrator, calibration_report

rng = np.random.default_rng(42)
latent = rng.normal(size=2000)
probability = 1 / (1 + np.exp(-latent))
scores = 1 / (1 + np.exp(-1.8 * latent))
y = rng.binomial(1, probability)

calibrator = CenteredIsotonicCalibrator().fit(scores[:1000], y[:1000])
predictions = calibrator.transform(scores[1000:])
report = calibration_report(y[1000:], predictions)
print(report)
```

Proper scores remain the default diagnostic because they reward both calibration and the information in a forecast (DeGroot and Fienberg, 1983; Gneiting and Raftery, 2007). Calibre reports the Brier score and its CORP decomposition into miscalibration, discrimination, and uncertainty, computed from an isotonic fit of outcomes on forecasts (Dimitriadis et al., 2021). It also reports binned, debiased, sweep, and smooth calibration errors (Kumar et al., 2019; Roelofs et al., 2022; Błasiok and Nakkiran, 2024), and the number of distinct forecast values. Distinct values measure granularity, not usefulness.

5 Evidence

5.1 With known risk, preserving distinctions can help or hurt

A finite population of 1,000 equally weighted score types, with five known risk functions, lets the two losses of Section 2 be computed exactly. Each of 30 calibration samples per design and sample size draws types with replacement and Bernoulli outcomes; every fitted map is then evaluated on the whole population. Appendix F gives the design.

When risk rises smoothly, isotonic regression's flat regions pool cases whose risks differ, and methods without them do better. At 1,000 calibration observations, centered isotonic regression lowers total

probability error relative to isotonic regression by -0.731 (95% interval [-0.830, -0.643]) on the squared-probability scale multiplied by 1,000, and the spline by -1.060 ([-1.186, -0.925]). Temperature scaling does better still, -2.045 ([-2.244, -1.858]), because this design lies inside its parametric family.

When risk is genuinely flat, the ranking reverses (Figure 3). In the step-risk design centered isotonic regression retains more risk information than isotonic regression but has larger total error at the middle sample size: estimating the extra probability levels costs more than the information is worth. With constant risk no map loses information, so any added variation is pure error. Interpolating across a flat region is an assumption that risk varies there, not a free recovery of information.

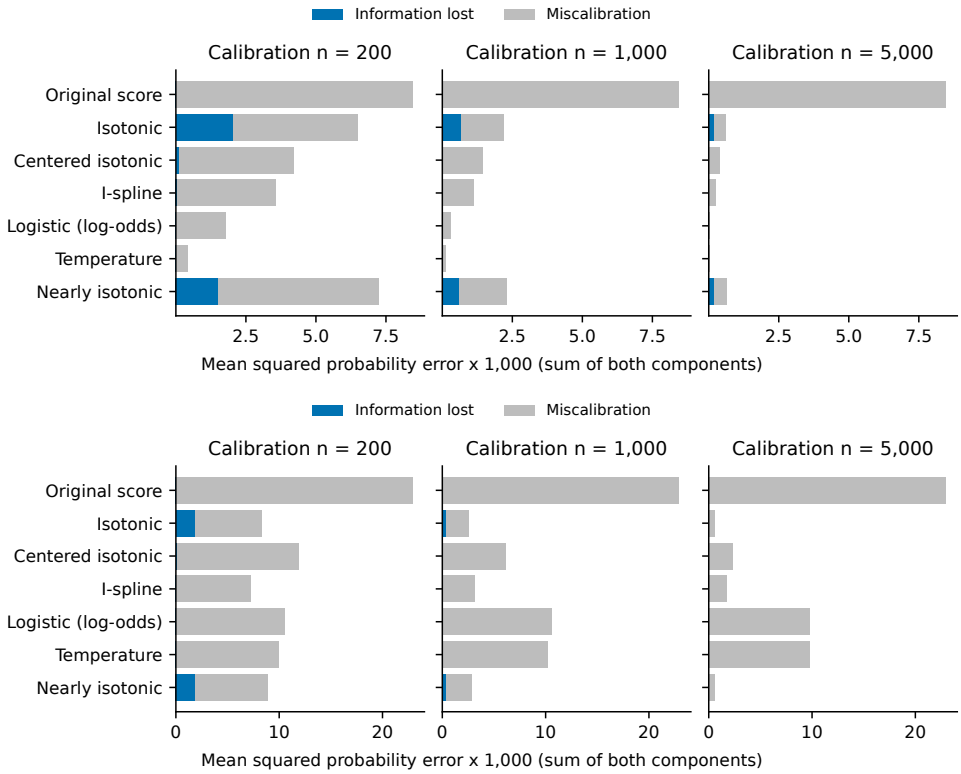


Figure 3: Total probability error split into information lost through pooling (blue) and miscalibration (gray), multiplied by 1,000, for smooth linear risk (top) and genuinely flat risk regions (bottom). Bars average 30 calibration samples per size over the complete 1,000-type population. Total error equals twice the uniformly integrated threshold regret. Axes are shared within each design. Paired intervals and threshold-specific curves are in Appendix F.

5.2 The benchmark, and what it shows about accuracy

The offline benchmark crosses 7 synthetic designs, which report a known distortion of a known probability, with the Wisconsin breast-cancer dataset scored by logistic regression, a random forest, and gradient boosting (Wolberg et al., 1993). That gives 10 configurations, 9 methods, and 30 seeds. Each seed holds out 40% of the data for testing. For real data, 5 stratified folds produce out-of-fold base-model scores for fitting the calibrator. Every calibrator in a configuration and seed sees identical inputs and runs at library defaults. Comparisons are paired within seed, with 2000 bootstrap resamples of the seed-level differences; real-data intervals describe split variability on one dataset, not uncertainty across datasets. Appendix G describes the designs.

On accuracy, the benchmark reproduces the known-risk pattern. In the overconfidence design centered isotonic regression keeps 1514 distinct test forecasts on average against 49 for isotonic regression, with mean Brier scores of 0.1527 and 0.1530 (Table 2). Temperature scaling is closest to the true probabilities, with mean absolute error 0.0040 against 0.0205 for centered isotonic regression, because the distortion is a pure temperature change. Leaving well-calibrated scores alone can also win: for logistic regression on the real data, the uncalibrated scores beat isotonic regression by 0.0013 in Brier score (interval $[0.0003, 0.0025]$).

Table 2: Overconfidence design, means over 30 seeds. Brier and MCB (miscalibration) are on the squared-probability scale; lower is better. Distinct counts forecasts after rounding to six decimals; more is not itself better. Truth MAE is mean absolute error against the generating probability. The duplicate isotonic wrapper is omitted.

	Brier	MCB	Distinct	Truth MAE
Uncalibrated	0.1604	0.0121	1594	0.0808
Isotonic	0.1530	0.0025	49	0.0255
Logistic (log-odds)	0.1521	0.0039	1599	0.0064
Temperature	0.1522	0.0039	1599	0.0040
Centered isotonic	0.1527	0.0044	1514	0.0205
I-spline	0.1524	0.0042	1595	0.0175
Nearly isotonic	0.1531	0.0025	52	0.0259
CDI-ISO	0.2982	0.0892	2	0.2999

5.3 Choosing a calibrator for the decision does not beat choosing it on Brier score

Before the first decision run, the benchmark configuration fixed 13 cost ratios from 0.02 to 0.98, 5 capacities, and one primary comparison. For each cost ratio, one procedure picks the calibrator with the highest validation payoff at that ratio; the other picks the calibrator with the lowest validation Brier score. Both choose among the same candidates, including act-all and act-none, using the same out-of-fold calibrated predictions on the calibration sample. The selection criterion is the only difference. The test half is scored once.

Payoff selection did not win (Figure 4). Of 130 configuration-by-threshold comparisons, 8 intervals lie above zero and 11 below. Chance alone would put about three on each side, so some thresholds show real differences, but they favor neither procedure, and no difference exceeds 2.7 per 1,000 cases. The largest losses are in the small-sample design, where a noisier selection criterion overfits the validation sample.

The reason is that there was little to win. In the synthetic designs, where the true probability is known, a rule that acts on it beats isotonic regression by a median of 0.47 per 1,000 cases across cost ratios (Table 3). The ceiling is higher only where flat regions straddle thresholds with real risk variation inside them: the small-sample design, whose isotonic fits are coarse, and the nonmonotone design at a cost ratio of 0.5, where its largest value is 7.6. A procedure tuned to the decision has to find gains of this size through the noise of a validation sample, and here it could not.

CDI-ISO, which builds the thresholds into the fit, did worse than either selector. It lost to isotonic regression in all 130 configuration-by-threshold cells, with every interval below zero, by a median of 29 and up to 206 per 1,000 cases. The cause is mechanical. With continuous scores almost every score value holds one observation, so the uncertainty bound that is meant to restrain decreases is wider than the unit interval, and the fit reproduces the training labels as zeros and ones. Its losses are smallest in the tied-score design, where scores repeat. Calibre now warns when most score groups are singletons.

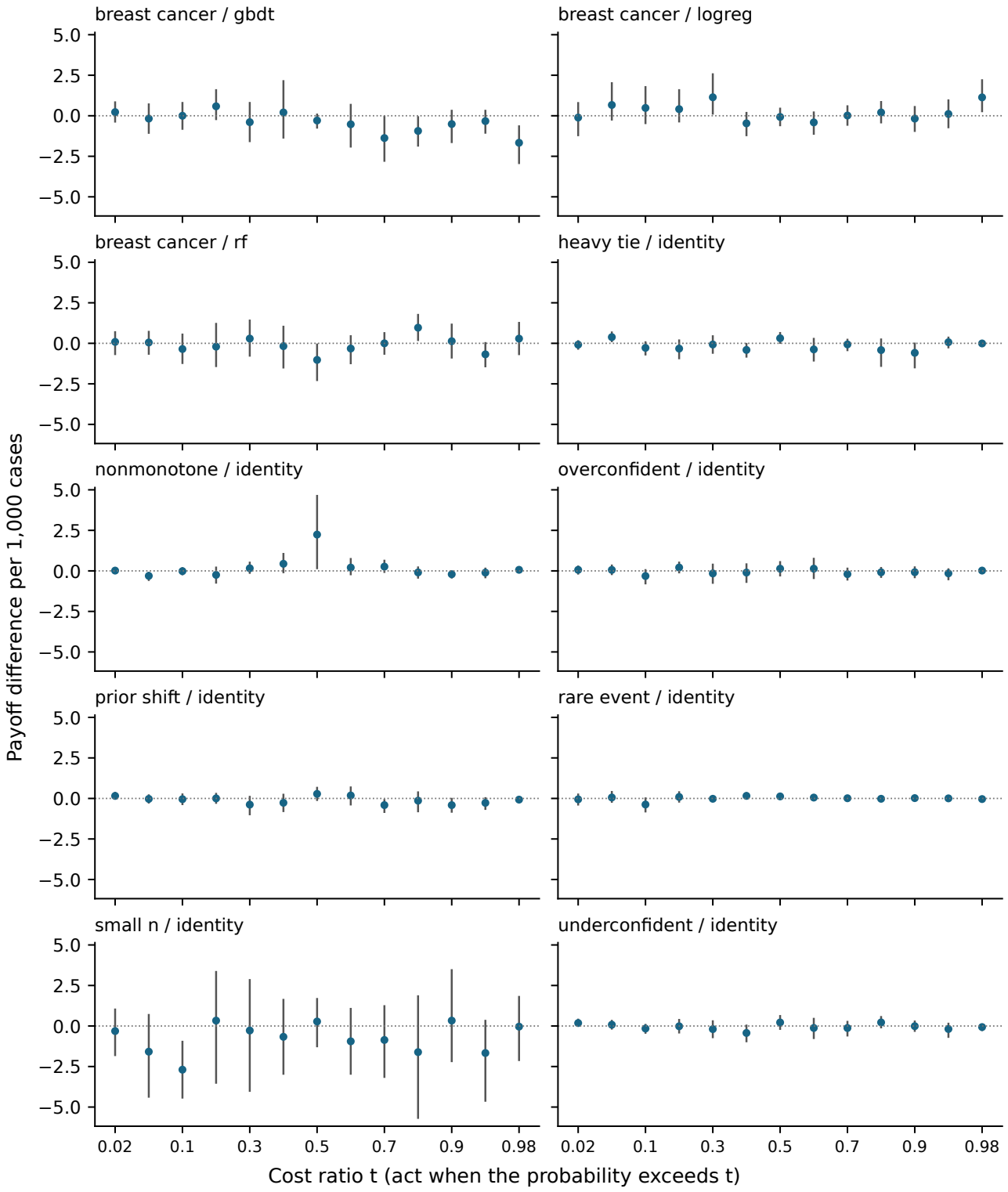


Figure 4: Test payoff of the calibrator chosen by validation payoff minus that of the calibrator chosen by validation Brier score, per 1,000 cases, at each pre-registered cost ratio. Positive values favor payoff selection. Dots are means over 30 paired seeds; lines are 95% percentile intervals from 2000 bootstrap resamples of the seed-level differences. Cost ratios are evenly spaced on the axis though not in value. All panels share both axes.

Table 3: Payoff of acting on the true probability minus payoff of acting on isotonic regression's probability, per 1,000 test cases, in the synthetic designs. Means over 30 seeds at selected cost ratios; every ratio is in the replication files. Small negative values occur because realized test outcomes are noisy.

Cost ratio	0.05	0.1	0.3	0.5	0.7	0.9
heavy tie	0.4	0.5	0.8	0.2	0.5	0.6
nonmonotone	0.2	0.3	0.6	7.6	0.3	0.4
overconfident	0.4	0.2	0.9	-0.1	0.5	0.5
prior shift	0.4	0.2	0.9	-0.1	0.5	0.5
rare event	0.2	0.6	0.2	0.1	0.1	0.2
small n	1.1	3.3	3.4	2.8	2.8	1.8
underconfident	0.4	0.2	0.9	-0.1	0.5	0.5

5.4 Under a capacity constraint, ties are the cost

For allocation, isotonic regression's flat regions cost exactly what their ties cost (Table 4). Filling 10% or 25% of slots by isotonic probability with random tie-breaking loses up to 1.7 successes per 1,000 cases relative to the original-score ranking. Breaking the same ties by the original score loses nothing in any configuration, as Proposition 4 requires; the benchmark checks this identity in every cell and stops if it fails. Centered isotonic regression, which has few ties, recovers most of the loss in the synthetic designs and little of it on the real data, where its anchors still tie the extreme scores.

Table 4: Expected successes selected at 10% and 25% capacity, minus those of the original-score ranking, per 1,000 test cases; means over 30 seeds. "Isotonic" and "Centered" break ties at random, with the expectation computed exactly; "score ties" breaks isotonic regression's ties by the original score.

	10% capacity			25% capacity		
	Isotonic	Centered	Isotonic, score ties	Isotonic	Centered	Isotonic, score ties
breast cancer / gbdt	-0.72	-0.72	0.00	-1.51	-1.42	0.00
breast cancer / logreg	-0.66	-0.66	0.00	-1.55	-1.55	0.00
breast cancer / rf	-0.57	-0.57	0.00	-1.43	-1.41	0.00
heavy tie / identity	-0.25	0.00	0.00	-0.55	0.00	0.00
nonmonotone / identity	-0.31	0.00	0.00	-0.07	0.00	0.00
overconfident / identity	-0.33	0.00	0.00	-0.57	0.00	0.00
prior shift / identity	-0.33	0.00	0.00	-0.57	0.00	0.00
rare event / identity	-0.38	0.00	0.00	-0.11	0.00	0.00
small n / identity	-0.37	-0.22	0.00	-0.12	0.00	0.00
underconfident / identity	-0.33	0.00	0.00	-0.57	0.00	0.00

The nonmonotone design is the exception that confirms the rule. There the original order is wrong, and at 50% capacity isotonic regression's pooling beats it; restoring the original order would forfeit that gain.

6 What to do

The evidence supports a short procedure. Choose a calibrator on a proper score on validation data, comparing the uncalibrated scores and a parametric correction as well as flexible maps. For allocation, break the calibrator's ties by the original score; this costs nothing when the original order is right. For cost-threshold decisions, check whether any fitted flat region straddles the thresholds in use and holds cases whose risks plausibly differ; only then is a map with fewer flat regions worth its estimation cost,

and the comparison should be made on held-out payoff at those thresholds. Do not tune for the decision on a small validation sample.

These conclusions have limits. Seven synthetic mechanisms and one real dataset cannot rank calibration methods in general, and repeated splits of one dataset are not independent datasets. The payoff model has one benefit, a known cost, and actions that do not change outcomes. A scalar monotone map cannot repair errors that vary across subgroups with the same score, and the cross-validation tools assume independent, exchangeable observations. The benchmark's spline fits are reproducible to the last few digits, not bit for bit: on near-separable data the optimizer stops at the limit of floating-point precision.

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Appendix A When replacement preserves a decision

Hold the calibration data and fitted map fixed. A fresh case has score S, binary state Y, and risk $p = P(Y = 1 \mid S)$. The probability-valued report is $Q = g(S) \in [0, 1]$ and $r = E[p \mid Q]$. All statements below concern this conditional evaluation population. They compare signals under a known population law; they do not presume that a deployed user knows that law. Actions do not change Y.

A decision maker chooses $a \in \{0, 1\}$ with payoff $a(Y - t)$ for a known cost ratio $t \in [0, 1]$. Let $V_t(S) = E[(p - t)_+]$ and $V_t(Q) = E[(r - t)_+]$ be optimal expected payoffs using each signal. The rule that treats the reported probability literally has payoff $U_t(Q) = E[(p - t)\mathbf{1}\{Q > t\}]$. Thus

$$V_t(S) - U_t(Q) = \underbrace{V_t(S) - V_t(Q)}_{D(t)\text{: replacement loss}} + \underbrace{V_t(Q) - U_t(Q)}_{\text{loss from using the reported level}}.$$

Both terms are nonnegative. This is a specialization of the value-of-information comparison, not a new decomposition principle (Blackwell, 1953; Tasche, 2021; Guo et al., 2026). The distinction between available information and its use also determines what a normative benchmark can establish (Hullman et al., 2025a).

A.1 A single threshold

Proposition 1 (crossing the decision boundary). Define

$$A_t(Q) = E[(p - t)_+ \mid Q], \qquad B_t(Q) = E[(t - p)_+ \mid Q].$$

Then $D(t) = E[\min\{A_t(Q), B_t(Q)\}]$. In particular, $D(t) = 0$ if and only if, for almost every report, its conditional risks do not put positive probability strictly on both sides of t.

Proof. Since $r - t = A_t - B_t$, the conditional payoff difference is $A_t - (A_t - B_t)_+ = \min(A_t, B_t)$. Its expectation vanishes precisely when the nonnegative minimum vanishes almost surely. The two conditional expectations are positive precisely when $P(p > t \mid Q) > 0$ and $P(p < t \mid Q) > 0$, respectively. Risks exactly at t are indifferent and do not create loss. □

This condition concerns actions, not the number of distinct probabilities. In the two-group example of Figure 1, pooling costs nothing when the cost is below both risks or above both risks. It costs value between them. A constant-risk population loses nothing under any pooling. Under arbitrary finite action sets, the same reasoning becomes a common-action condition: in a finite population, each positive-mass report group must admit an action optimal for every score type in that group. This follows because the expected nonnegative regret of a common action is zero exactly when its regret is zero at each positive-mass type.

A.2 A range of decisions and direct use of probabilities

Proposition 2 (replacement over a threshold range). On a finite score population with positive type weights, let m_q and M_q be the smallest and largest risks assigned report q. For any specified threshold set $T \subseteq [0,1]$, replacement preserves optimal payoff at every $t \in T$ if and only if

$$(m_q, M_q) \cap T = \varnothing \qquad \text{for every report group } q.$$

For all thresholds in $[0,1]$, replacement is lossless if and only if $p = r$ almost surely. This last equivalence also holds for general score distributions.

Proof. In a finite group, both strict sides of t carry positive mass exactly when $m_q < t < M_q$. Apply Proposition 1. For the general all-threshold statement, the identity $2\int_0^1 D(t)dt = E[(p-r)^2]$ proves necessity, and $p = r$ proves sufficiency. □

An average requires a declared distribution of decisions. For a probability measure W on thresholds, define $\int D(t)\, dW(t)$. This is zero exactly when replacement is lossless for W-almost every threshold. Since $D(t)$ is continuous, zero average also implies zero loss throughout the support of W, including its non-atomic thresholds. It need not establish safety outside that support. Uniform weighting recovers half the squared grouping loss; a different weighting describes a different economic comparison. Threshold mixtures and decision curves are established tools (Ehm et al., 2016; Vickers and Elkin, 2006).

Proposition 3 (using Q as a probability). Relative to the original-score oracle, the rule $\mathbf{1}\{Q > t\}$ has regret

$$R(t) = E\left[|p - t|\, \mathbf{1}\{\mathbf{1}\{Q > t\} \neq \mathbf{1}\{p > t\}\}\right].$$

It is optimal at t precisely when $P(p > t, Q \leq t) = P(p < t, Q > t) = 0$. In particular, a report equal to t declines the action and can be costly when $p > t$. It is optimal at every threshold if and only if $Q = p$ almost surely. By contrast, it is optimal among rules using only Q at every threshold if and only if $Q = r$ almost surely.

Proof. Subtract the chosen payoff from $(p-t)_+$ separately for $p > t$ and $p \leq t$. Differences at $p = t$ have zero cost. Integrating gives $2\int R(t)dt = E[(p-Q)^2]$. Conditioning on Q gives the same argument with r in place of p, and integrated loss $E[(r-Q)^2]/2$. □

Calibration ($Q = r$) therefore makes the report directly usable relative to its own information. Replacing the score without loss requires $p = r$; combining both requirements gives $Q = p$. Hullman's discussion of calibration and additional information motivates making these benchmarks explicit (Hullman, 2024; Hullman et al., 2025b). Calibration decision loss measures improvement obtainable by recalibrating predictions over bounded decision problems (Hu and Wu, 2024). Replacement loss instead asks what was discarded before that correction is possible. A constant calibrated report can have zero calibration decision loss and positive replacement loss.

Appendix B Retaining a score and being able to use it

B.1 Capacity allocation

Proposition 4 (recovering the original allocation). For a fixed batch of cases and a nondecreasing map g, sorting first by descending $Q = g(S)$ and then by descending S produces the same order as sorting by descending S. Resolve any original-score ties by the same fixed rule in both procedures. The selected cases then agree at every capacity, including fractional selection at a boundary.

Proof. If $S_i > S_j$, monotonicity gives $Q_i \geq Q_j$. A strict report inequality orders the pair correctly; a report equality is resolved by $S_i > S_j$. If the original scores tie, the common tie rule applies. □

This is an allocation consequence of an established tie-handling strategy (Menon et al., 2012). It establishes equality with the original ranking, not superiority to a risk oracle or even to random selection. For equal-value, equal-capacity cases, ranking by S is optimal for every capacity when $p(S)$ is nondecreasing. A nonmonotone risk function can make restoring that order harmful. Variable benefits or resource requirements require a different allocation rule.

B.2 Available information versus recoverable payoff

Because Q is a fixed function of S, the pair (Q, S) has the same information about Y as S. Hence $V_t(Q, S) = V_t(S)$. Publishing both preserves information relative to the score, but does not teach a user the unknown function $p(S)$. The fitted calibrator can help a user choose actions even though, conditional on the map, it adds no information to an oracle who already knows the population law.

If g is invertible and its inverse is available, any rule learned from scored examples can be transported to Q: decode the examples and the new report, then run the original procedure. It has exactly the same finite-sample payoff. This is a correspondence between procedures, not a claim that every fixed learner is invariant to a change of coordinates. An advantage found by giving the two encodings different hypothesis classes, training budgets, or access to the map must be attributed to those restrictions.

For pooled reports, decoding is impossible without additional information. For rounded reports the relevant map is $S \mapsto \text{round}(g(S))$, even if g itself is invertible. For someone with extra context Z, the appropriate comparison is $V_t(S, Z) - V_t(Q, Z)$, using $P(Y = 1 \mid S, Z)$ and $P(Y = 1 \mid Q, Z)$. The present simulation has no Z and cannot establish effects on expert decisions. The dependence of information value on available signals is explicit in Guo et al. (2026); the difference between idealized and actual uses of prediction information is central to Hullman et al. (2025b).

A subsequent recoverability study would have to fix the permitted learner, labeled data available to each report recipient, access to the calibration map, and reporting precision before comparing payoffs. Its baseline must include probability plus score. The current experiments evaluate literal threshold rules and specified ranking rules; they do not estimate how much additional data a user needs to learn to exploit a retained score.

Appendix C Contribution assessment

The established results above supply a precise test of the package's motivation: retained distinctions matter when they permit different useful actions. They do not establish a new theorem about calibration or information. Table C1 records the status of the claims instead of treating their appearance in this paper as evidence of novelty.

Table C1: Claims, precedents, and what this paper establishes. "Specialization" means a derivation for the stated score-replacement problem, with no novelty claim.

Claim	Closest foundation	Status here
Pooling reduces available decision value	Blackwell (1953); Tasche (2021)	Established; explicit signal comparison.
Pooling matters at crossed thresholds	Jensen equality and grouping loss	Specialization: Propositions 1–2.
Calibration supports literal use within its signal	Hu and Wu (2024); Hullman (2024)	Established; Proposition 3 separates two benchmarks.
Original-score ties restore original ranks	Menon et al. (2012)	Established strategy; capacity consequence in Proposition 4.
Value depends on information and its use	Hullman et al. (2025); Guo et al. (2026)	Established framing; no human behavior measured.
Threshold curves compare useful decisions	Ehm et al. (2016); Vickers and Elkin (2006)	Established evaluation tools.
Payoff recoverable from finite labeled data	Known-inverse equivalence; specified learner needed	Candidate research question; no new bound or estimator established.

The propositions are consequences of existing theory, and original-score tie-breaking is prior art. What the paper adds is their assembly into a test a practitioner can run, the software that runs it, and the evidence on how large the costs are in practice.

A stronger contribution would be a finite-sample characterization of when retaining the original score yields an exploitable payoff gain over a pooled report, after accounting for estimation error and reporting precision. That question remains open. Answering it needs an explicit class of risk functions and feasible learning procedures, and a result the precedents above do not supply; more method-by-design simulations would not settle it.

The known-risk simulation was run before the decision framing was written, and the offline decision evaluation was pre-registered in the benchmark configuration before its first run. Every design and method is reported.

Appendix D Adjacent increment constraints

For sorted unique score groups with $W_j > 0$, consider

$$\min_{z \in \mathbb{R}^m} \sum_j W_j(\bar{y}_j - z_j)^2 \quad \text{subject to} \quad z_{j+1} - z_j \geq L_j.$$

Set $R_1 = 0$, $R_{j+1} = R_j + L_j$, and $u_j = z_j - R_j$. The constraints become $u_{j+1} \geq u_j$ and the loss is $\sum_j W_j(\bar{y}_j - R_j - u_j)^2$, so one weighted isotonic projection gives $z_j = u_j + R_j$. This is the total-order case of margin-separated isotonic projection (Gunasekar et al., 2016).

The projection takes $O(m)$ time once scores are sorted and ties aggregated. The unbounded problem is feasible for any finite L_j, but positive bounds can require cumulative increases larger than one.

Calibre solves the unbounded problem and optionally clips to $[0, 1]$; clipping preserves nonpositive bounds but can flatten positive ones, so the clipped output is not generally the constrained optimum with a probability box.

Appendix E Cost- and data-informed isotonic calibration

For normalized threshold weights a_k and bandwidth h, CDI-ISO's economic weight between adjacent unique scores is

$$e_j = \sum_k a_k \max\{0, 1 - |\tfrac{1}{2}(x_j + x_{j+1}) - t_k|/h\}.$$

Within each unique-score group the Kish effective size is $n_j^{\text{eff}} = (\sum_i w_i)^2 / \sum_i w_i^2$. Adjacent groups define a pooled rate $\tilde{p}_j$ weighted by these sizes, and

$$q_j = \sqrt{\tilde{p}_j(1 - \tilde{p}_j)\left(1/n_j^{\text{eff}} + 1/n_{j+1}^{\text{eff}}\right)}.$$

With $c = \Phi^{-1}(1 - \alpha/2)$, the bound is

$$L_j = \gamma e_j[\bar{y}_{j+1} - \bar{y}_j - cq_j]_+ - (1 - e_j)cq_j,$$

and the cumulative-shift solver of Appendix D fits a right-continuous step map. Thresholds and scores must be on the same probability scale.

When every group has one observation, q_j is roughly $\sqrt{2\tilde{p}_j(1 - \tilde{p}_j)}$, so cq_j exceeds one for most event rates. Away from the thresholds the bound then permits any decrease, and near them the positive part almost never binds: the fit reproduces the labels. That is the failure Section 5.3 measures. The construction needs repeated scores, such as rounded or binned ones, and its thresholds, bandwidth, and multiplier need application-specific validation.

Appendix F Known-risk simulation

The population is 1,000 equally weighted types $s_j = (j + 0.5)/1000$ with reported scores $\text{logistic}(1.8\,\text{logit}(s_j))$. True risks follow five designs: s; $0.05 + 0.9s^2$; levels 0.2, 0.5, and 0.8 on equal thirds; constant 0.3; and $0.5 + 0.35\sin(2\pi s)$. The first admits an exact temperature correction and the last violates monotone ordering. Calibration samples of 200, 1,000, and 5,000 draw types with replacement and independent Bernoulli outcomes, 30 seeds per design and size, and all seven methods see the same sample. Evaluation integrates over the whole population, computing r over exact output groups. Intervals use 2,000 bootstrap resamples of the 30 seeds, paired across methods; they are pointwise 95% intervals for mean performance across calibration samples. There is no evaluation-label noise. A six-decimal rounding sensitivity, which can only increase pooling loss, is in the replication files.

For capacity, each fraction from 1% to 99% selects the largest outputs and computes expected successes per 1,000 population members, with random boundary ties averaged exactly or broken by the original score (Figure F1).

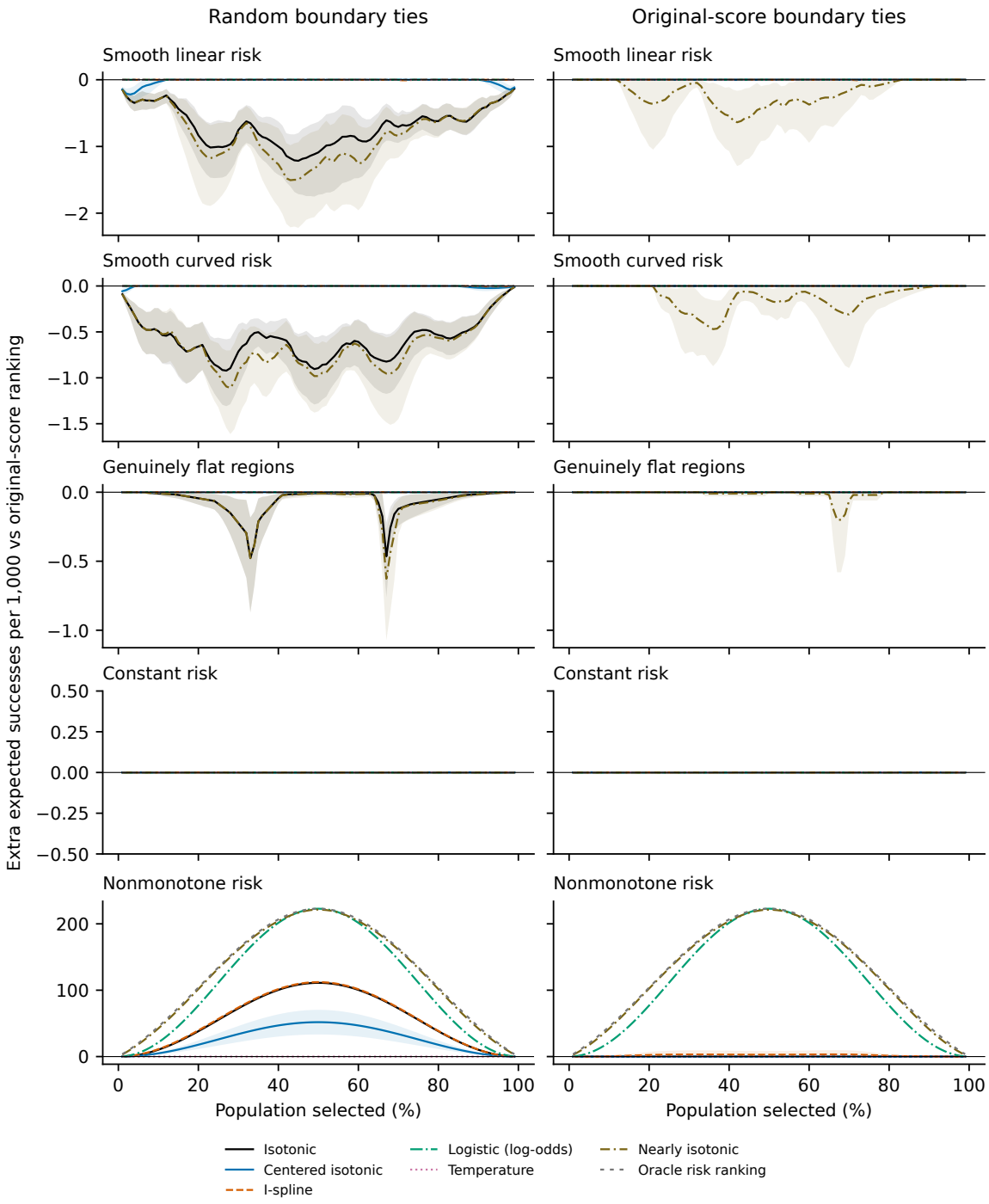


Figure F1: Capacity allocation in the known-risk simulation at calibration size 1,000: extra expected successes per 1,000 population members relative to original-score ranking. Left: random boundary ties, averaged exactly. Right: ties broken by the original score. The oracle ranks by true risk. Shading is a paired 95% pointwise interval for the mean across calibration samples. Each row shares an axis; constant-risk differences are zero to floating-point precision.

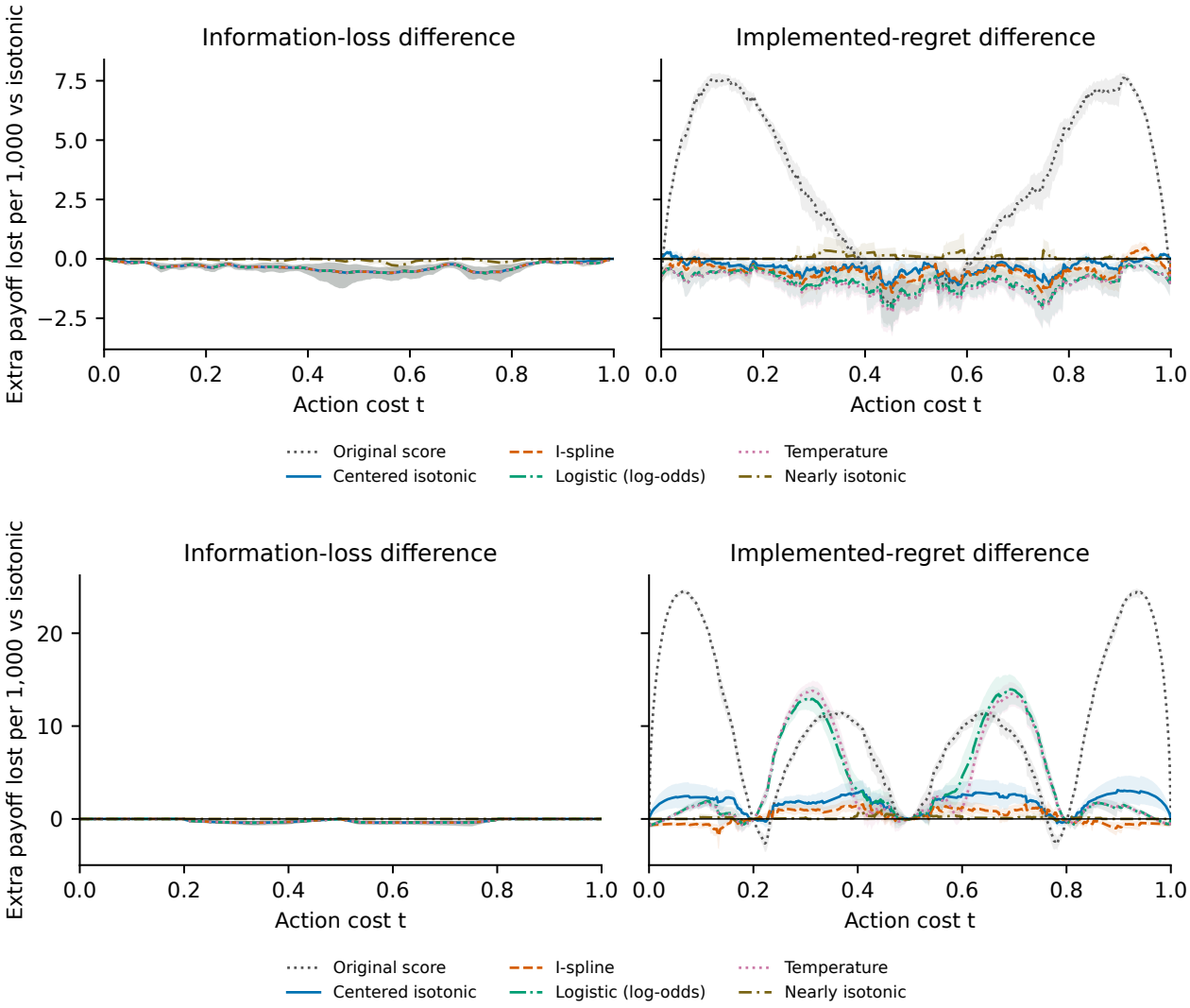


Figure F2: Threshold-specific comparisons with isotonic regression at calibration size 1,000, for smooth linear risk (top) and genuinely flat risk regions (bottom). Left panels: difference in optimal payoff lost to pooling. Right panels: difference in regret from acting when the reported probability exceeds the cost ratio. Negative values favor the named method. Shading gives paired 95% pointwise percentile intervals from 2,000 seed resamples.

Appendix G Offline benchmark

In the overconfident, underconfident, and prior-offset designs, $Z \sim N(0, 2^2)$, the true probability is $\sigma(Z)$, and the reported scores are $\sigma(1.8Z)$, $\sigma(0.6Z)$, and $\sigma(Z + 0.8)$. The tied-score design rounds the overconfident score to two decimals. The rare-event design draws $Z \sim N(-4.6, 1.5^2)$ and reports $\sigma(1.6Z)$. The small-sample design uses the overconfidence construction with 300 observations instead of 4,000. The nonmonotone design draws $s \sim U(0, 1)$ and clips $s + 0.25 \sin(2\pi s)$ to $[0.01, 0.99]$. Outcomes are Bernoulli. Three designs share one latent score and differ only by strictly increasing transformations, so rank-based methods, including isotonic regression and the oracle, give identical results across them.

Comparators are the uncalibrated scores, isotonic regression, logistic regression on the log-odds, and temperature scaling fitted by bounded scalar optimization; none uses scikit-learn's `CalibratedClassifierCV`. Spline and nearly isotonic defaults include internal cross-validation, so tuning budgets differ. CDI-ISO receives the pre-registered cost ratios as its thresholds, a problem input rather than tuning, and library defaults otherwise. For the selection comparison, each candidate's out-of-fold predictions on the calibration sample come from 5 folds. Every benchmark cell also checks that Calibre's isotonic wrapper matches scikit-learn's to 10^{-12} and that original-score tie-breaking restores the original allocation for every non-decreasing map.

Table G1: Complete offline benchmark means over 30 seeds. Counts are rounded to integers. – marks truth error, which real data cannot supply. Paired Brier intervals are in Figure G1.

		Brier	MCB	Log loss	AUC	Distinct	Truth MAE	Seconds
breast cancer / gbdt	Uncalibrated	0.0307	0.0082	0.1431	0.9903	191	–	< 0.001
	Isotonic	0.0316	0.0058	0.2886	0.9865	12	–	0.002
	Logistic (log-odds)	0.0295	0.0070	0.1129	0.9903	226	–	0.005
	Temperature	0.0294	0.0069	0.1119	0.9903	227	–	0.002
	Calibre isotonic	0.0316	0.0058	0.2886	0.9865	12	–	0.002
	Centered isotonic	0.0311	0.0081	0.2938	0.9871	78	–	0.002
	I-spline	0.0306	0.0081	0.1403	0.9903	198	–	1.817
	Nearly isotonic	0.0325	0.0066	0.4036	0.9824	20	–	0.087
	CDI-ISO	0.0579	0.0049	2.0867	0.9413	2	–	0.002
breast cancer / logreg	Uncalibrated	0.0208	0.0058	0.0817	0.9941	196	–	< 0.001
	Isotonic	0.0221	0.0035	0.2965	0.9892	9	–	0.002
	Logistic (log-odds)	0.0209	0.0060	0.0900	0.9941	167	–	0.004
	Temperature	0.0208	0.0059	0.0896	0.9941	171	–	0.002
	Calibre isotonic	0.0221	0.0035	0.2965	0.9892	9	–	0.003
	Centered isotonic	0.0216	0.0050	0.2949	0.9898	56	–	0.002
	I-spline	0.0206	0.0056	0.1006	0.9941	167	–	1.791
	Nearly isotonic	0.0244	0.0044	0.4592	0.9840	13	–	0.068
	CDI-ISO	0.0384	0.0029	1.3858	0.9614	2	–	0.002
breast cancer / rf	Uncalibrated	0.0346	0.0083	0.1587	0.9882	80	–	< 0.001
	Isotonic	0.0355	0.0058	0.3043	0.9843	10	–	< 0.001
	Logistic (log-odds)	0.0335	0.0073	0.1592	0.9882	80	–	0.005
	Temperature	0.0332	0.0069	0.1576	0.9882	80	–	0.001
	Calibre isotonic	0.0355	0.0058	0.3043	0.9843	10	–	0.002
	Centered isotonic	0.0343	0.0077	0.3009	0.9851	48	–	< 0.001
	I-spline	0.0343	0.0077	0.1424	0.9878	76	–	1.693
	Nearly isotonic	0.0358	0.0062	0.3050	0.9841	11	–	0.101
	CDI-ISO	0.0575	0.0061	1.9393	0.9388	4	–	0.003
heavy tie / identity	Uncalibrated	0.1604	0.0111	0.6380	0.8603	101	0.0810	< 0.001

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Table G1: Complete offline benchmark means over 30 seeds. Counts are rounded to integers. – marks truth error, which real data cannot supply. Paired Brier intervals are in Figure G1.

		Brier	MCB	Log loss	AUC	Distinct	Truth MAE	Seconds
	Isotonic	0.1532	0.0022	0.4685	0.8590	22	0.0265	0.001
	Logistic (log-odds)	0.1558	0.0066	0.5142	0.8603	101	0.0524	0.003
	Temperature	0.1558	0.0066	0.5142	0.8603	101	0.0523	0.001
	Calibre isotonic	0.1532	0.0022	0.4685	0.8590	22	0.0265	0.001
	Centered isotonic	0.1528	0.0035	0.4673	0.8603	101	0.0216	0.001
	I-spline	0.1525	0.0033	0.4665	0.8603	100	0.0189	1.610
	Nearly isotonic	0.1532	0.0023	0.4685	0.8588	23	0.0267	0.118
	CDI-ISO	0.1611	0.0051	0.5174	0.8458	71	0.0660	0.001
nonmonotone / identity	Uncalibrated	0.2457	0.0339	0.6921	0.6754	1598	0.1589	< 0.001
	Isotonic	0.2160	0.0022	0.6347	0.6820	37	0.0354	0.001
	Logistic (log-odds)	0.2224	0.0106	0.6304	0.6754	1598	0.0792	0.003
	Temperature	0.2224	0.0106	0.6306	0.6754	1598	0.0792	0.001
	Calibre isotonic	0.2160	0.0022	0.6347	0.6820	37	0.0354	0.001
	Centered isotonic	0.2165	0.0047	0.6358	0.6754	1520	0.0405	0.006
	I-spline	0.2156	0.0038	0.6145	0.6793	1164	0.0304	1.697
	Nearly isotonic	0.2157	0.0025	0.6341	0.6905	45	0.0310	0.789
	CDI-ISO	0.4284	0.1837	15.4403	0.5716	2	0.4278	0.003
overconfident / identity	Uncalibrated	0.1604	0.0121	0.5187	0.8606	1594	0.0808	< 0.001
	Isotonic	0.1530	0.0025	0.4891	0.8593	49	0.0255	0.002
	Logistic (log-odds)	0.1521	0.0039	0.4643	0.8606	1599	0.0064	0.004
	Temperature	0.1522	0.0039	0.4643	0.8606	1599	0.0040	0.002
	Calibre isotonic	0.1530	0.0025	0.4891	0.8593	49	0.0255	0.002
	Centered isotonic	0.1527	0.0044	0.4885	0.8606	1514	0.0205	0.008
	I-spline	0.1524	0.0042	0.4660	0.8606	1595	0.0175	3.218
	Nearly isotonic	0.1531	0.0025	0.4892	0.8590	52	0.0259	1.151
	CDI-ISO	0.2982	0.0892	10.7468	0.7018	2	0.2999	0.007
prior shift / identity	Uncalibrated	0.1696	0.0213	0.5104	0.8606	1599	0.1190	< 0.001
	Isotonic	0.1530	0.0025	0.4891	0.8593	49	0.0255	0.001
	Logistic (log-odds)	0.1521	0.0039	0.4643	0.8606	1599	0.0064	0.003
	Temperature	0.1675	0.0193	0.5025	0.8606	1599	0.1077	0.002
	Calibre isotonic	0.1530	0.0025	0.4891	0.8593	49	0.0255	0.001
	Centered isotonic	0.1527	0.0044	0.4882	0.8606	1514	0.0199	0.006
	I-spline	0.1524	0.0041	0.4662	0.8606	1596	0.0166	2.336
	Nearly isotonic	0.1531	0.0025	0.4892	0.8590	52	0.0259	0.770
	CDI-ISO	0.2982	0.0892	10.7464	0.7019	2	0.2999	0.003
rare event / identity	Uncalibrated	0.0245	0.0019	0.1220	0.8390	1144	0.0185	< 0.001
	Isotonic	0.0239	0.0009	0.1236	0.8317	23	0.0076	0.001
	Logistic (log-odds)	0.0236	0.0011	0.0991	0.8390	1558	0.0027	0.004
	Temperature	0.0236	0.0011	0.0988	0.8390	1560	0.0017	0.001
	Calibre isotonic	0.0239	0.0009	0.1236	0.8317	23	0.0076	0.001
	Centered isotonic	0.0238	0.0012	0.1231	0.8375	1189	0.0061	0.005
	I-spline	0.0238	0.0013	0.1009	0.8390	1407	0.0067	1.647
	Nearly isotonic	0.0240	0.0009	0.1240	0.8265	30	0.0083	0.775
	CDI-ISO	0.0481	0.0225	1.7336	0.5459	2	0.0479	0.004
small n / identity	Uncalibrated	0.1576	0.0263	0.4996	0.8644	120	0.0810	< 0.001
	Isotonic	0.1560	0.0144	0.7758	0.8573	16	0.0640	< 0.001
	Logistic (log-odds)	0.1509	0.0196	0.4597	0.8644	120	0.0283	0.002
	Temperature	0.1503	0.0190	0.4578	0.8644	120	0.0187	< 0.001

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Table G1: Complete offline benchmark means over 30 seeds. Counts are rounded to integers. – marks truth error, which real data cannot supply. Paired Brier intervals are in Figure G1.

		Brier	MCB	Log loss	AUC	Distinct	Truth MAE	Seconds
	Calibre isotonic	0.1560	0.0144	0.7758	0.8573	16	0.0640	< 0.001
	Centered isotonic	0.1543	0.0225	0.7752	0.8635	96	0.0579	< 0.001
	I-spline	0.1533	0.0220	0.5036	0.8642	117	0.0430	0.724
	Nearly isotonic	0.1568	0.0150	0.7786	0.8537	19	0.0675	0.057
	CDI-ISO	0.2992	0.0936	10.7819	0.6992	2	0.3002	< 0.001
underconfident / identity	Uncalibrated	0.1589	0.0107	0.4878	0.8606	1599	0.0772	< 0.001
	Isotonic	0.1530	0.0025	0.4891	0.8593	49	0.0255	0.002
	Logistic (log-odds)	0.1521	0.0039	0.4643	0.8606	1599	0.0064	0.004
	Temperature	0.1522	0.0039	0.4643	0.8606	1599	0.0040	0.001
	Calibre isotonic	0.1530	0.0025	0.4891	0.8593	49	0.0255	0.002
	Centered isotonic	0.1527	0.0044	0.4881	0.8606	1514	0.0198	0.007
	I-spline	0.1524	0.0041	0.4663	0.8606	1595	0.0166	3.505
	Nearly isotonic	0.1531	0.0025	0.4892	0.8590	52	0.0259	1.009
	CDI-ISO	0.2982	0.0892	10.7467	0.7018	2	0.2999	0.005

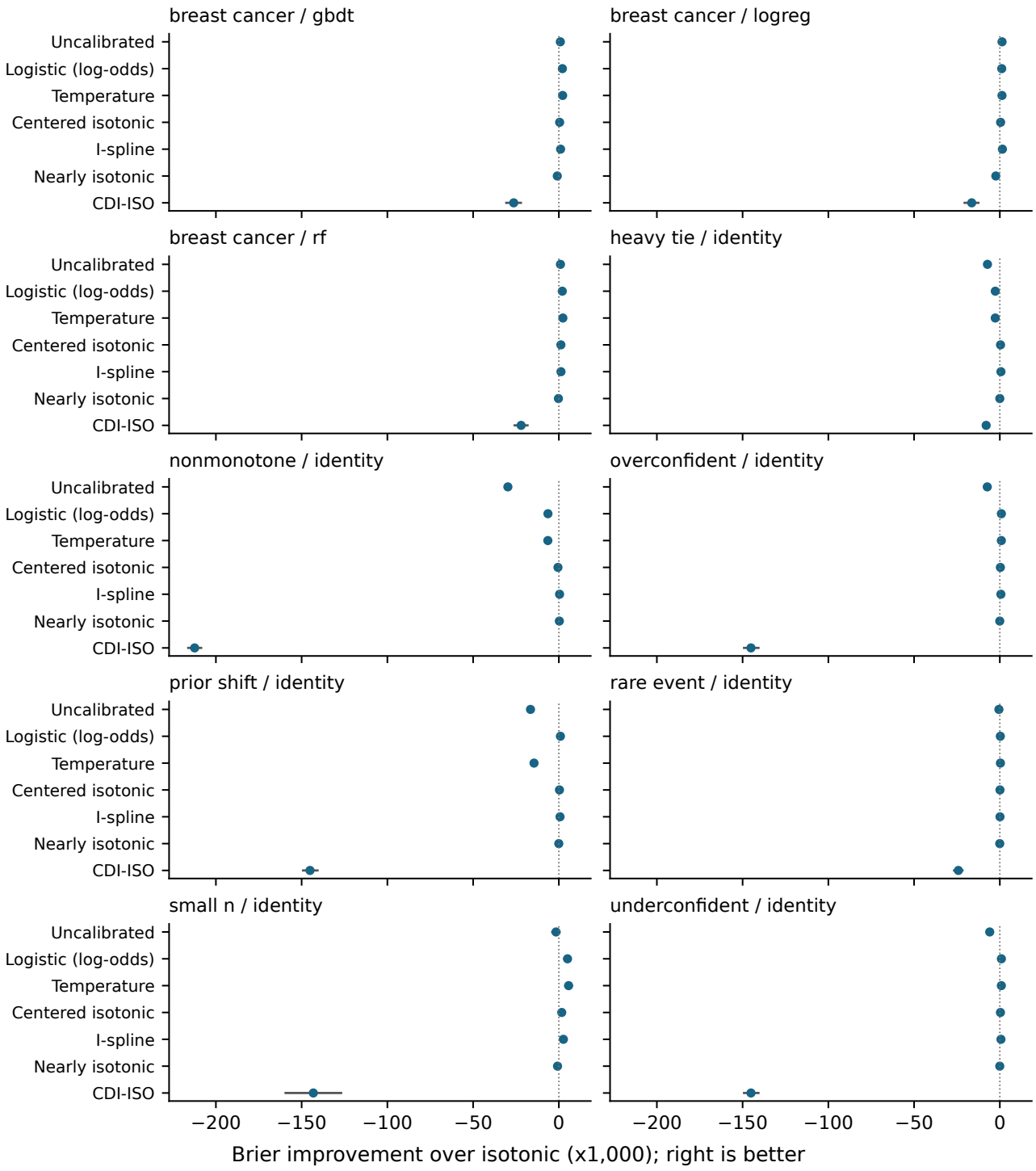


Figure G1: Paired Brier improvements over isotonic regression, multiplied by 1,000, for every configuration. Dots are mean differences over 30 seeds; lines are 95% percentile intervals from 2000 resamples of the seed-level differences. Positive values favor the named method. All panels share an axis.

Appendix H Reproduction

The package, benchmark, and simulation are at `https://github.com/finite-sample/calibre` under the MIT license, with documentation at `https://finite-sample.github.io/calibre/`. The following commands install the locked environment, rerun the benchmark and the simulation, and build this paper:

```
uv sync --locked --all-groups
uv run python -m benchmarks.run --n-jobs 4
uv run python -m benchmarks.aggregate
make granularity-run
make paper
```

`make paper` regenerates every exhibit and number from the committed results without refitting, and writes `ms/build/calibre.pdf`. Manifests record input hashes, source hashes, and the software environment. The pre-registered choices live in `benchmarks/config.py`.